Algorithms for Decoding Interstellar Messages

Michael Matessa
100 Pine St, Suite 1250, San Francisco, CA 94111
METI International
mmatessa@gmail.com

Abstract
How would we determine the meaning of an interstellar message from a distant civilization? This paper describes algorithms for assigning meaning to symbols found in such a message. It is assumed the message is pre-parsed, resulting in symbols grouped into expressions. Algorithms assign meanings to symbols within an expression and meta-algorithms maintain a list of symbol meanings across expressions. The algorithms described in this paper are Equality, Counting, Function, Base, Approximation, Constants, and Patterns. The meta-algorithms are Intersect, Prune, and Substitute. These algorithms have been implemented in software, and results from a sample message are given that assign meanings ranging from numbers to the concepts of mass and length. By developing algorithms, it is hoped that message decoding can grow from an art done by individuals to a science done with algorithms.

Keywords: Search for Extraterrestrial Intelligence, SETI, Interstellar messages, Decoding, Algorithms

1. Introduction

How would we determine the meaning of an interstellar message from a distant civilization? This paper describes algorithms for assigning meaning to symbols found in such a message. These algorithms have been implemented in software, and results from a sample message are given. By developing algorithms, it is hoped that message decoding can develop from an art into a science.

What previous work has been done on message decoding algorithms? Most algorithmic work has focused on detecting language-like structures [1, 2], but not decoding meaning. Non-algorithmic decoding strategies have been proposed that describe the process for analyzing a message [3, 4]. Also, constructed messages have been given to volunteers to study their decoding process [5-8].

The algorithms in this paper take advantage of several simplifying assumptions. Messages are assumed to consist of a linear sequence of symbols grouped into expressions. In a full decoding of a message, the symbols would need to be isolated from a stream of sub-symbolic constructors such as two frequencies representing binary values, but this work is assumed to have already been done. It is assumed that messages are constructed in a way that facilitates decoding: simple concepts are presented first, and more complex concepts are built on earlier parts of the message. Messages are assumed to refer to concepts in mathematics and physics that would be similar for all civilizations in our universe. And it is assumed that the whole message has been collected from its beginning without errors in transcription.

2. Studies on Decoding

In 1977, Carl Sagan gave his graduate students a long string of 0's and 1's. Without any further guidance, he asked his students to see if they could decode the message in 48 hours. There were 29791 0's and 1's,

which they soon figured out could be divided into a cube of 31x31x31. They tried to make a movie out of it, but that proved to be uninteresting. Finally, they settled on a three-dimensional image, which looked like a formaldehyde molecule. This molecule has a particular radio frequency associated with it. The students discovered what Carl Sagan had intended – tune to that frequency to find the rest of the message [5].

Michael Busch provided a message to five undergraduates and a colleague [6]. The undergrads each spent no more than an hour attempting to decode it, working without any pattern-recognition software. Three of the students correctly identified the delimiters '(' and ')', with two also identifying symbols representing '=', '≠', and the notation for integers. With about twelve hours of work, the colleague was able to correctly interpret the contents of the message [9].

A message decoding experiment was carried out by Alexander Ollongren [7]. One of Ollongren's graduate students produced formal annotations to a text of 74 sentences in the Czech language. The text was supplemented with logical descriptions of the objects and relations between them in terms of the Coq proof assistant. Ollongren, without knowing Czech or the meaning of the text, attempted to interpret the logic contents from the annotations. Using the logical descriptions, "much understanding of the textual contents could be achieved – but the text was far from completely understood" [7].

René Heller posted a constructed message to social media [8]. Within a month after the posting, over 300 replies were received in total, 66 of which contained the correct solutions. About half of the solutions were derived fully independently, the other half profited from public online discussions and spoilers.

From these examples we see that it is possible to discover meaning from a message without previously knowing the language that was used for its construction. However, the methods used for decoding were usually not reported.

3. Strategies for Decoding

Although not defining any algorithms, Brian McConnell [3] and Michael Matessa [4] describe general strategies for decoding interstellar messages. McConnell's steps are to

1. Understand the mode and modulations of the message
2. Understand how basic units of information are formulated
3. Understand symbol frequency and structure
4. Parse the message

McConnell states that in the third step "we will use a variety of techniques, such as entropy analysis, to study how the communication stream is structured". In the last step "we will be able to use similar statistical techniques to parse this language, to identify the most frequently used 'words' and the concepts that are most interlinked with others, and zero in on these for further investigation" [3].

The steps described by Matessa are to

1. Find the dimension of the message
2. Find the symbols in the message
3. Find the symbol meanings

Matessa describes features that can help in determining meaning in the third step of finding symbol meanings [4]. These features include sub-symbolic type, redundant symbols, expression consistency, physics ratios, and physics expression patterns. In a sub-symbolic representation, the initial bits of a symbol may be used to represent the symbol type (e.g., number or math function), or frequencies can be used to "color code" the type of the symbol. Redundant symbols give extra context in an expression. For example, instead of using a unique symbol for "seconds", the concepts can be represented with symbols for "time" and "SI unit". Although it may not be necessary for decoding, this redundancy can help make connections to meanings in other parts of the message. By introducing new symbols in logically consistent expressions, new symbol meanings may be provided in new contexts. Unit-less physics ratios do not depend on knowing unit definitions such as pound or kilogram. For example, the ratio of the mass of a proton to the mass of an electron is approximately 1836 in any unit system, and this ratio has been used in several transmitted messages [6, 10]. Finally, physics expression patterns are a feature that can help map symbols to meanings. For example, the Planck unit system is defined in terms of universal physical constants in unique patterns. These patterns can be used by algorithms to determine symbol meaning, as described in the next section.

4. Algorithms for Decoding

Algorithms have been used for creating interstellar messages and detecting them, but not for decoding them (assigning meanings to symbols). For example, Alexander Ollongren used an automated proof assistant to check the correctness of logical expressions in a created message [11]. Also, John Elliott developed a language-detection algorithm "without necessarily having to 'understand' the content" [1]. The algorithm uses a number of statistical clues: data compression ratio, "chunking" to find character bit-length and boundaries, and matching against a Zipfian type-token distribution for "letters" and "words" [1].

This paper describes algorithms for assigning meanings to symbols in an expression and meta-algorithms for maintaining a list of symbol meanings across expressions. As described in the introduction, the algorithms take advantage of several simplifying assumptions. Messages are assumed to consist of a linear sequence of symbols, grouped into expressions. The algorithms described in this paper are Equality, Counting, Function, Base, Approximation, Constants, and Patterns. The meta-algorithms are Intersect, Prune, and Substitute.

4.1 Equality

Mathematical equality is a powerful function that enables the presentation of new symbol meanings and their equivalence to previously established symbol meanings. For example, once the symbols for numbers have been established, symbols for mathematical functions such as addition and multiplication can be introduced.

The Equality algorithm uses symbol patterns in expressions showing the reflexive property to determine what symbol maps onto the equals sign. The equals sign may be at the beginning, middle, or end of the expression, representing prefix, infix, and postfix notation respectively. For example, after seeing the expressions "s01 s10 s01" and "s03 s10 s03", the algorithm maps s10 to "=" and places the assumption that the notation is infix in a hypothesis list.

4.2 Counting
Natural numbers are easily described with multiple objects. The Counting algorithm uses patterns of repeating symbols to assign number values to symbols. The algorithm also matches patterns for one symbol or zero symbols.

4.3 Function
The meanings of symbols for mathematical functions are deduced by the Function algorithm by determining which functions are logically consistent with the numbers found in the expression. For example, in the expression "0 s14 0 = 0", s14 could potentially be "+", "-", or "*". The algorithm determines these potential meanings for binary operators by iterating through a list of operators and calculating which are logically consistent. The algorithm then places these in a hypothesis list to compare with future expressions. As with equality, the symbol for the function may be at the beginning, middle, or end of the expression, representing prefix, infix, and postfix notation respectively.

4.4 Base
Although humans tend to use base ten for mathematics, cultures have used other bases, such as the Babylonians using base sixty. An alien species with two hands and six fingers on each (the Twelve-toes) might use base twelve. The Base algorithm searches for place value system patterns defining the base of the mathematics in the message. For example, if the algorithm saw "5 + 5 = 1 0" it would note the base as ten in a hypothesis list, but if it saw a message from the Twelve-toes society with "6 + 6 = 1 0" it would note the base as twelve.

The Base algorithm also computes the value of multidigit numbers using the base. In a message from the Twelve-toes with twelve as the hypothesized base, in the expression "1 2 s17 7 = 2" the algorithm would read "1 2" as the decimal number 12+2=14. Similarly, in the expression "1 s17 2 = 0 . 6" from the Twelve-toes, the algorithm would read "0 . 6" as the decimal number 6/12=0.5.

4.5 Approximation
Approximation is a useful concept for describing repeating numbers, irrational numbers, and physics ratios. The Approximation algorithm looks for patterns in expressions that can be used to define an approximation symbol. For example, in "1 / 3 s21 0 . 3 3 3" the algorithm would map the symbol s21 to to the meaning "approximately" in a hypothesis list.

4.5 Constants
Dimensionless quantities such as pi and e have the same value in any culture and can be used in pattern matching of more complex physics equations. The Constants algorithm searches for known values to assign to symbols. For example, in " s22 approx 3 . 1 4 1 5 9" the algorithm would assign the symbol s22 the meaning of "pi".

One way to bridge mathematics and physics in an interstellar message is to use unitless physics ratio constants, which do not depend on knowing unit definitions such as pound or kilogram. For example, the ratio of the mass of a proton to the mass of an electron is approximately 1836, and the ratio of the neutron mass to the electron mass is approximately 1839. The Constants algorithm looks for ratio patterns to define the meanings of the symbols. Given the expression "s23 [s24] / s23 [s25] approx 1 8 3 6" the Constants algorithm (plus other meta-algorithms) would assign s23 the meaning of "mass",

s24 the meaning of “proton”, and s25 the meaning of “electron”. The fine-structure constant α is another unitless quantity defined by the ratio of universal physical constants, as seen in Figure 1.

$$\alpha = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{\hbar c} \approx \frac{1}{137}$$

Fig. 1: The fine-structure constant

Given the expression “s26 = [s27 expt 2] / [4 * pi * s28 * s29 * c]” the algorithm would assign s27 the meaning of “e”, s28 the possible meanings of “e0”or “h-bar”, and s29 the possible meanings of “e0”or “h-bar”.

4.6 Patterns

Another way to discuss measurements in a message is to use a system of measurement that does not depend on previously defined Earth standards. One such system is the Planck unit system. Planck units are defined exclusively in terms of four universal physical constants: c, the speed of light in a vacuum; G, the gravitational constant; h-bar, the reduced Planck constant; and k_B, the Boltzmann constant. Figure 2 gives the definitions of Planck length, Planck mass, Planck time, and Planck energy:

Name	Expression
Planck length	$l_\mathrm{P} = \sqrt{\frac{\hbar G}{c^3}}$
Planck mass	$m_\mathrm{P} = \sqrt{\frac{\hbar c}{G}}$
Planck time	$t_\mathrm{P} = \sqrt{\frac{\hbar G}{c^5}}$
Planck energy	$E_\mathrm{P} = \sqrt{\frac{\hbar c^5}{G}}$

Fig. 2: Planck unit expressions

The Planck unit system provides patterns that can be used to determine the meanings of individual symbols. The Planck algorithm is given expressions with symbols for basic mathematics that have been previously decoded, such as

s31 [s32] = sqrt [[s29 * s33] / [s15 ^ 3]]

and adds meanings for symbols consistent with the pattern into hypothesis lists. For example, after given the expression above the hypothesis lists would contain s31=(planck, length), s32=(planck, length), s29=(h-bar, G), s33=(h-bar, G), and s15=(c). Given the expression

s31 [s23] = sqrt [[s29 * s15] / s33]

the hypothesis lists would contain s31=(planck, mass), s32=(planck, mass), s29=(h-bar, c), s15=(h-bar, c), and s33=(G). Meta-algorithms derive logical consistency between these hypotheses and give hypothesis lists of s31=(planck), s32=(length), s23=(mass), s29=(h-bar), s33=(G), and s15=(c).

4.7 Intersect

In addition to algorithms that place symbol meanings into hypothesis lists, there are also meta-algorithms that manage the logical consistency of the hypothesis lists. The Intersect algorithm is invoked when hypotheses are added. It updates the list with the intersection of the added hypotheses and previous entries in the hypothesis list. For example, after the expression "0 s15 0 = 0", the hypothesis list of s15=(+ , - , *). The subsequent expression "4 s15 2 = 2" has a hypothesis list of s15=(- , /) in isolation, but the Intersect algorithm finds that the logically consistent hypothesis list from both expressions is the intersection of the isolated hypotheses, and so s15=(-).

4.8 Prune

If a symbol has a single hypothesis in its hypothesis list, the Prune algorithm removes that hypothesis from the hypothesis lists of other symbols. For example, after the expression " s23 [s24] / [s23 [s25]] approx 1 8 3 6" the Constants algorithm adds the hypothesis lists of s23=(mass, proton), s24=(mass, proton), s23=(mass, electron), and s25=(mass, electron). The Intersect algorithm updates the hypothesis list of s23 to s23=(mass). The Prune algorithm then updates the hypothesis lists of s24 and s25 to s24=(proton) and s25=(electron).

4.9 Substitute

If a symbol has a single hypothesis, the Substitute algorithm substitutes that hypothesis for the symbol in the whole message. For example, after the expressions "s01 = s11 s12 s13" and "s02 = s11 s12 s12 s13" the hypothesis list of s01=(1) and s02=(2). Later in the message, Substitute algorithm allows the expression "s01 s14 s01 = s02" to be interpreted as "1 s14 1 = 2".

5. Implemented System

The algorithms described above have been implemented in the Lisp programming language and applied to a test message. The test message is shown in Appendix A. It consists of a series of symbols (labelled s1 through s33) that are grouped into expressions. The meanings of these symbols range from numbers to concepts of mass and length. The algorithms are able to read in these symbols and produce a decoded message, shown in Appendix B. The mappings of symbols to meanings determined by the algorithms are shown in Appendix C.

6. Conclusions

This paper presents algorithms that can be used to determine the meaning of an interstellar message from a distant civilization. This is only an initial step in developing decoding algorithms. The algorithm pattern matching is currently limited and could be improved. For example, it assumes the use of the constant pi but other systems may use tau which is equal to two times pi. Likewise, it assumes the use of h-bar, the reduced Planck constant, and not the Planck constant itself, which is equal to 2*pi*h-bar. Also, the pattern matching in the algorithms does not include other potentially relevant expressions such

as $E=mc^2$, $L=L_0*\sqrt{(1-v^2/c^2)}$, or $\mu_0 \varepsilon_0=1/c^2$. A more robust pattern matching would include these variations and others. But it is hoped that this effort is the start of improving message decoding from an art done by individuals to a science done with algorithms.

In addition to decoding, the algorithms could also help in the creation of messages. Pre- and post-conditions of the individual algorithms could be chained backwards to determine a series of supporting concepts to discuss a target concept. For example, to discuss the concept "proton", the Constants algorithm indicates that an expression relating the ratio of the mass of the proton and the mass of the election to 1836 is needed. To discuss ratios, the Function algorithm indicates that expressions relating the division symbol to integers numbers are needed, and so on. Describing this chain forward, a message could be created with these supporting concepts.

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Appendix A: Test Message

(s01 s10 s01)
(s02 s10 s02)
(s03 s10 s03)
(s01 s10 s11 s12 s13)
(s02 s10 s11 s12 s12 s13)
(s03 s10 s11 s12 s12 s12 s13)
(s04 s10 s11 s12 s12 s12 s12 s13)
(s05 s10 s11 s12 s12 s12 s12 s12 s13)
(s06 s10 s11 s12 s12 s12 s12 s12 s12 s13)
(s07 s10 s11 s12 s12 s12 s12 s12 s12 s12 s13)
(s08 s10 s11 s12 s12 s12 s12 s12 s12 s12 s12 s13)
(s09 s10 s11 s12 s12 s12 s12 s12 s12 s12 s12 s12 s13)
(s00 s10 s11 s13)
(s04 s14 s05 s10 s09)
(s09 s15 s04 s10 s05)
(s02 s16 s03 s10 s06)
(s06 s17 s02 s10 s03)
(s02 s18 s03 s10 s08)
(s11 s01 s14 s03 s13 s10 s04)
(s02 s16 s11 s01 s14 s03 s13 s10 s08)
(s09 s17 s11 s01 s14 s02 s13 s10 s03)
(s19 s11 s04 s13 s10 s02)
(s19 s11 s09 s13 s10 s03)
(s08 s14 s01 s10 s09)
(s09 s14 s01 s10 s01 s00)
(s09 s14 s02 s10 s01 s01)
(s01 s00 s16 s01 s00 s10 s01 s00 s00)
(s00 s20 s01 s10 s01 s17 s01 s00)
(s00 s20 s00 s01 s10 s01 s17 s01 s00 s00)
(s03 s20 s01 s10 s03 s14 s01 s17 s01 s00)
(s01 s17 s03 s21 s00 s20 s03 s03 s03)
(s22 s21 s03 s20 s01 s04 s01 s05 s09)
(s23 s11 s24 s13 s17 s11 s23 s11 s25 s13 s13 s21 s01 s08 s03 s06)
(s26 s21 s01 s17 s01 s03 s07)
(s26 s10 s11 s27 s18 s02 s13 s17 s11 s04 s16 s22 s16 s28 s16 s29 s16 s30 s13)
(s31 s11 s32 s13 s10 s19 s11 s11 s29 s16 s33 s13 s17 s11 s30 s18 s03 s13 s13)
(s31 s11 s23 s13 s10 s19 s11 s11 s29 s16 s30 s13 s17 s33 s13)

Appendix B: Message Decoded by Algorithms

(1 = 1)
(2 = 2)
(3 = 3)
(1 = [o])
(2 = [o o])
(3 = [o o o])
(4 = [o o o o])
(5 = [o o o o o])
(6 = [o o o o o o])
(7 = [o o o o o o o])
(8 = [o o o o o o o o])
(9 = [o o o o o o o o o])
(0 = [ ])
(4 + 5 = 9)
(9 - 4 = 5)
(2 * 3 = 6)
(6 / 2 = 3)
(2 expt 3 = 8)
([1 + 3] = 4)
(2 * [1 + 3] = 8)
(9 / [1 + 2] = 3)
(sqrt [4] = 2)
(sqrt [9] = 3)
(8 + 1 = 9)
(9 + 1 = 1 0)
(9 + 2 = 1 1)
(1 0 * 1 0 = 1 0 0)
(0 . 1 = 1 / 1 0)
(0 . 0 1 = 1 / 1 0 0)
(3 . 1 = 3 + 1 / 1 0)
(1 / 3 approx 0 . 3 3 3)
(pi approx 3 . 1 4 1 5 9)
(mass [proton] / [mass [electron]] approx 1 8 3 6)
(alpha approx 1 / 1 3 7)
(alpha = [e expt 2] / [4 * pi * e0 * h-bar * c])
(planck [length] = sqrt [[h-bar * G] / [c expt 3]])
(planck [mass] = sqrt [[h-bar * c] / G])

Appendix C: Symbol-Meaning Mappings

s00 (0)
s01 (1)
s02 (2)
s03 (3)
s04 (4)
s05 (5)
s06 (6)
s07 (7)
s08 (8)
s09 (9)
s10 (=)
s11 ([)
s12 (o)
s13 (])
s14 (+)
s15 (-)
s16 (*)
s17 (/)
s18 (expt)
s19 (sqrt)
s20 (.)
s21 (approx)
s22 (pi)
s23 (mass)
s24 (proton)
s25 (electron)
s26 (alpha)
s27 (e)
s28 (e0)
s29 (h-bar)
s30 (c)
s31 (planck)
s32 (length)
s33 (G)